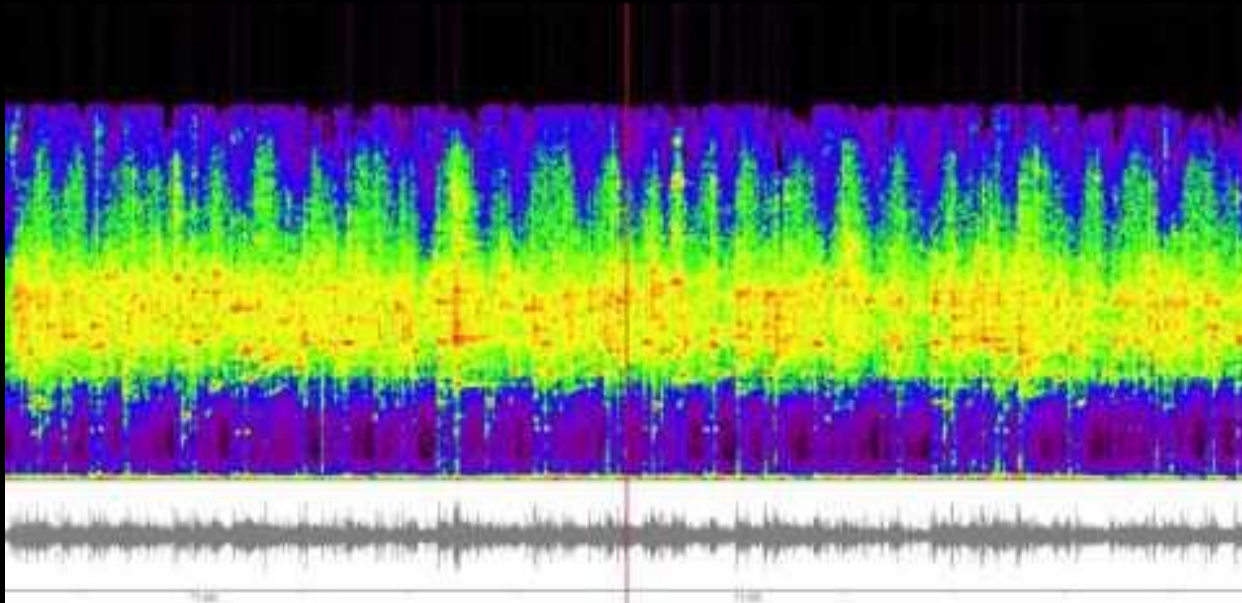


Granular Synthesis

Anxe Faraldo

anxefaraldo@gmail.com

Iannis Xenakis: *Concret PH* (1958)



The Sound Grain

Dennis Gabor (1947) referred to sound grains as acoustical quanta.

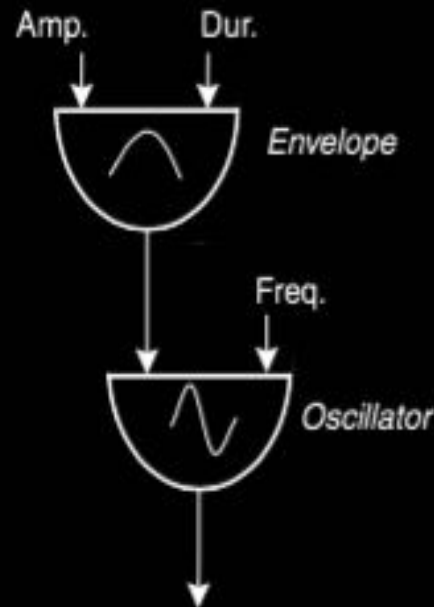
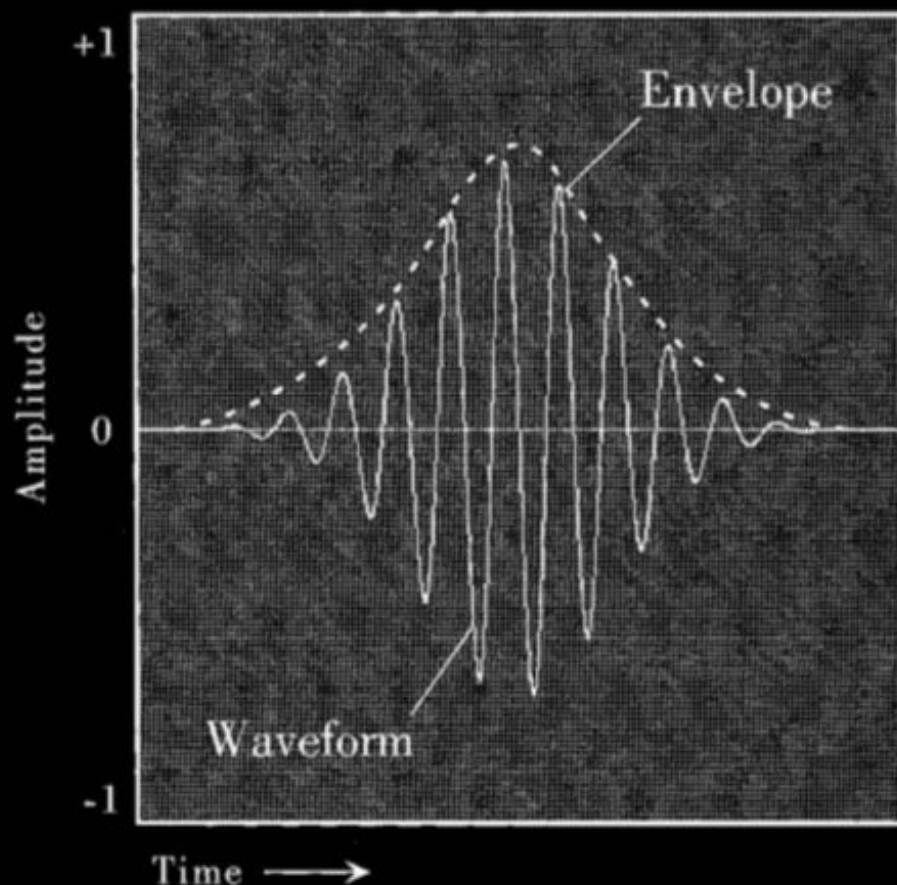


Figure 5.8: A granular synthesis instrument may be as simple as a single oscillator and a Gaussian-like bell-shaped envelope

Gabor, D. (1947) Acoustical quanta and the theory of hearing, *Nature*, 159(4044). Figure taken from Reck Miranda, E. (2002). *Computer Sound Design*. Focal Press.

The Sound Grain



“Fourier analysis is a timeless description in terms of exactly periodic waves of infinite duration. On the other hand it is our most elementary experience that sound has a time pattern as well as a frequency pattern. A mathematical description is wanted which takes account of this duality.”

Gabor 1947, p. 591

Figure 3.1 Portrait of a grain in the time domain. The duration of the grain is typically between 1 and 100 ms.

“All sound is an integration of grains...”

All sound is an integration of grains, of elementary sonic particles, of sonic quanta. Each of these elementary grains has a threefold nature: duration, frequency, and intensity. All sound, even all continuous sonic variation, is conceived as an assemblage of a large number of elementary grains adequately disposed in time. So every sonic complex can be analyzed as a series of pure sinusoidal sounds even if the variations of these sinusoidal sounds are infinitely close, short, and complex. In the attack, body, and decline of a complex sound, thousands of pure sounds appear in a more or less short interval of time. Hecatombs of pure sounds are necessary for the creation of a complex sound.

Iannis Xenakis, *Formalized Music* (p. 43)

The Sound of Granular Synthesis

Xenakis: Analogique A-B (1959)

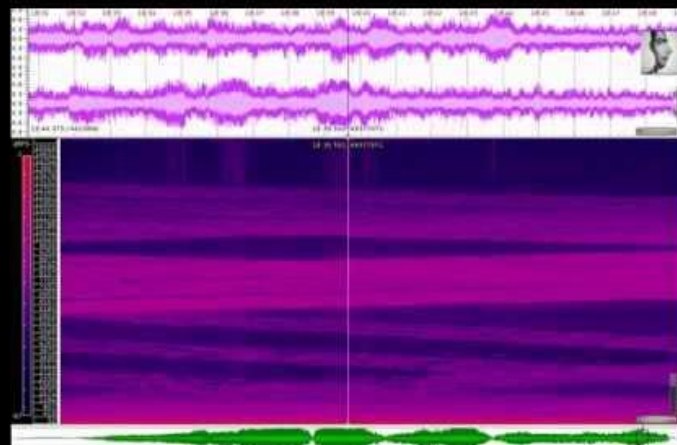
1st ever granular piece

Splicing tape manually!



Barry Truax: Riverrun (1986)

1st realtime digital piece



<http://www.sfu.ca/~truax/river.html>

Granular Sampling

The technique of *micromontage* is an essential component of Vaggione's style. In micromontage, the composer extracts particles from sound files and rearranges them in time and space. The term 'montage' derives from the world of cinema where it refers to cutting, splicing, dissolving and other film editing operations. The term 'micro' refers to the manner in which a composer can position each sound particle precisely on the canvas of time. 'Digital micromontage' refers to operations dealing with small sound particles, belonging to the micro-time domain (usually less than 100 ms).

Horacio Vaggione: Schall (1994)

Types of Granular Synthesis

Synchronous Granular Synthesis

Time Stretching, VoSim, etc.

grains triggered at regular time intervals to produce a sound with a particular pitch period

Asynchronous Granular Synthesis

Clouds and Textures

grains are played asynchronously to produce a "cloud" of sound

grains are taken from one or many soundfiles or sound sources (waveforms)

envelopes are crucial to express the desired qualities of the sound

Synchronous Granular Synthesis: VoSim

VoSim = Voice Simulation ~ Emulates Vocalic Formants

Series of $(\sin)^2$ pulses followed by silence

Control Parameters:

N pulses per period (N)

Duration of each pulse (T)

Delay between pulse groups (M)

Amplitude of 1st pulse (A)

Decay slope

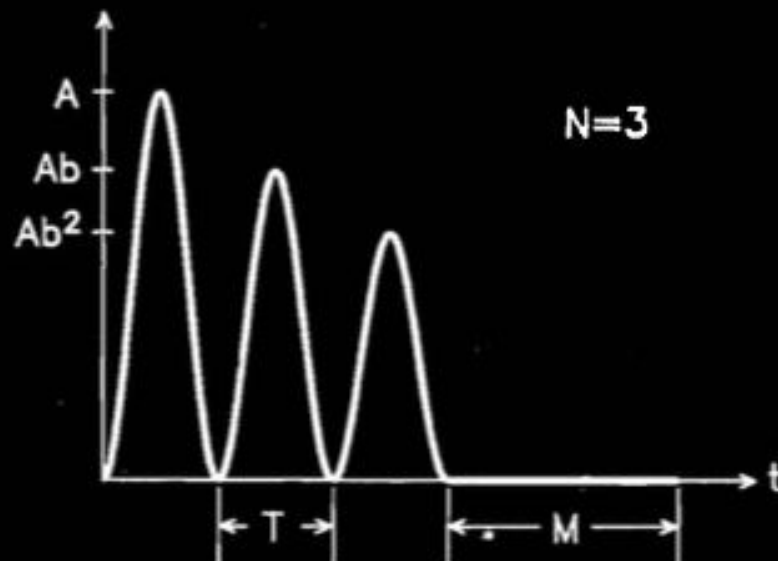


FIGURE 8.6 Basic VOSIM waveform.

Synchronous Granular Synthesis: VoSim



Asynchronous Granular Synthesis: Parameters

Grain duration

Grain density

Inter-onset time interval

Frequency bands / boundaries

Amplitude envelope

Waveform

Spatial dispersion

Asynchronous Granular Synthesis: Control Models

One-to-one

One-to-many

A single control may affect various parameters simultaneously mapped

According to different transfer functions, scales, offsets, etc.

Tendency masks (from G. M. Koenig)

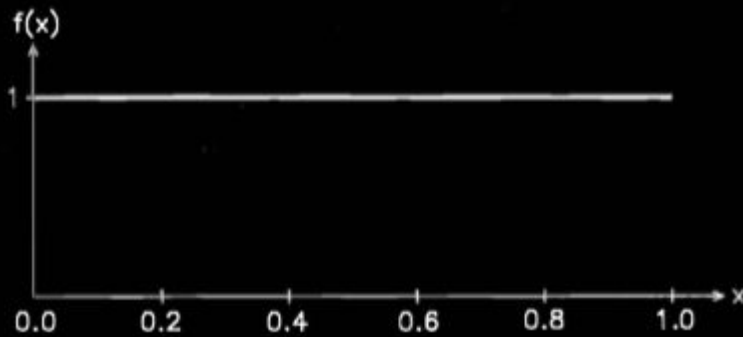
Parameters move randomly between dynamic boundaries

Probabilistic distributions

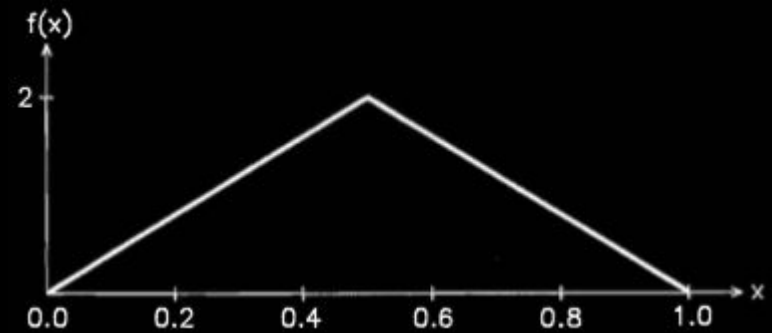
Expressed as means and variances

Common Probability Distributions

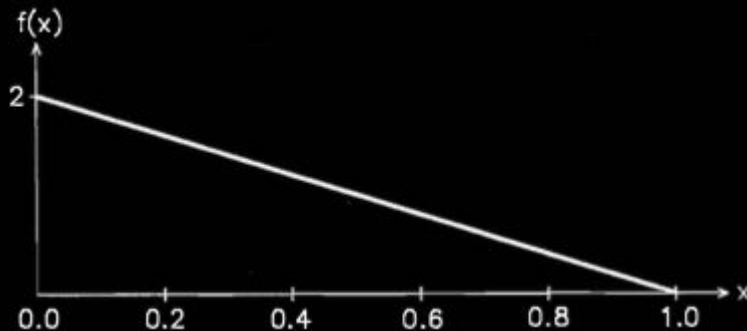
Uniform



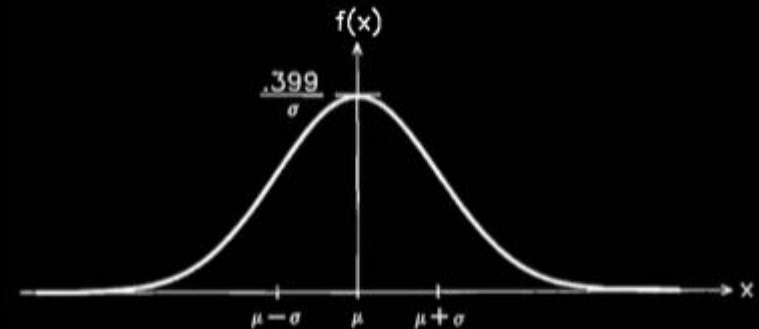
Triangular



Linear

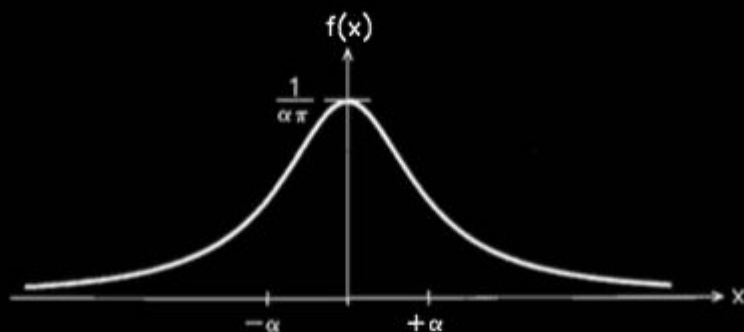


Gaussian

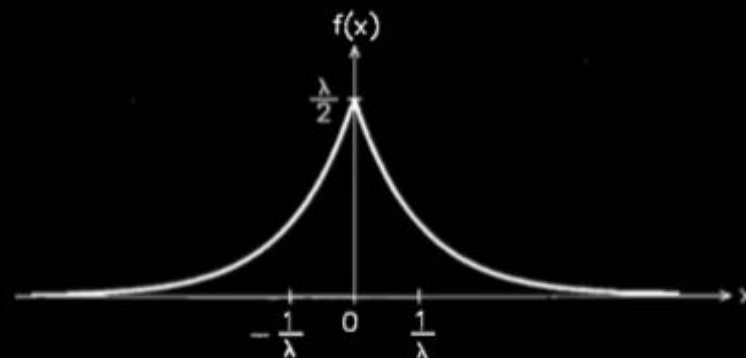


Common Probability Distributions

Cauchy



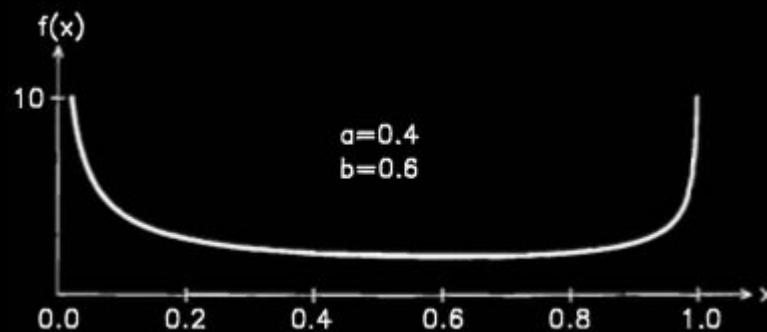
Bilinear exponential



Exponential

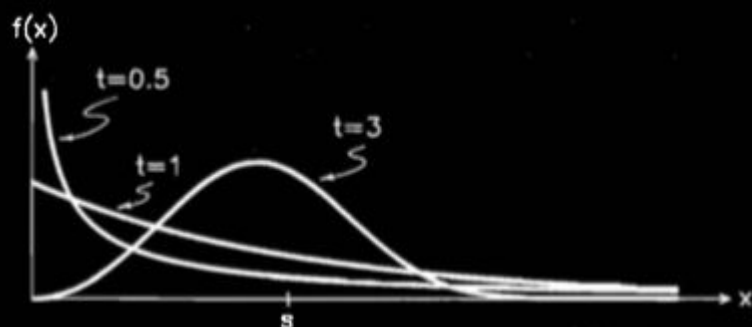


Beta

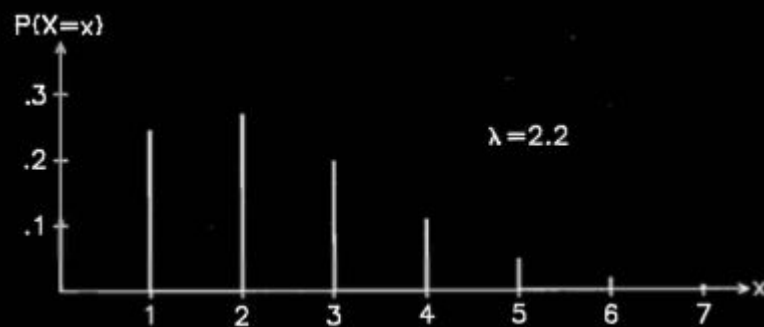


Common Probability Distributions

Weibull



Poisson



In Pd, these distributions are implemented in 'maxlib', but they can be easily implemented as abstractions.

It might be interesting to explore how different they 'sound'

Available Granular Synthesis Implementations

Anxe Faraldo (Pd):

<https://github.com/anxefaraldo/pd-grainders>

Olivier Belanger (python, standalone):

<http://ajaxsoundstudio.com/software/soundgrain/>

Ross Bencina (standalone):

<http://www.audiomulch.com/>

and maaaaaaany more!

<http://granularsynthesis.com/software.php>

Essential Bibliography

Xenakis, Iannis (1971). *Formalized Music*. Pendragon Press.

Roads, Curtis (2001). *Microsound*. MIT Press.

Bencina, Ross (2001). Implementing Real-time Granular Synthesis. in Greenbaum & Barzel (eds.), *Audio Anecdotes III*. A.K. Peters.

Roads, Curtis (1988). *Introduction To Granular Synthesis*.
Computer Music Journal, 12(2).

<http://granularsynthesis.com/hthesis/hthesis.html>

Summary: Implementation Fundamentals

Basic building block:

many (waveform * envelope) > 100 ms.

The creation of **multiple instances** of the same grain

In Pd, create and abuse abstractions

Techniques such as dynamic patching can come in handy

The control of randomness and probability is essential

Shape global behaviour rather than individual grains